CS60111 GIS · TERM PROJECT · GROUP 8

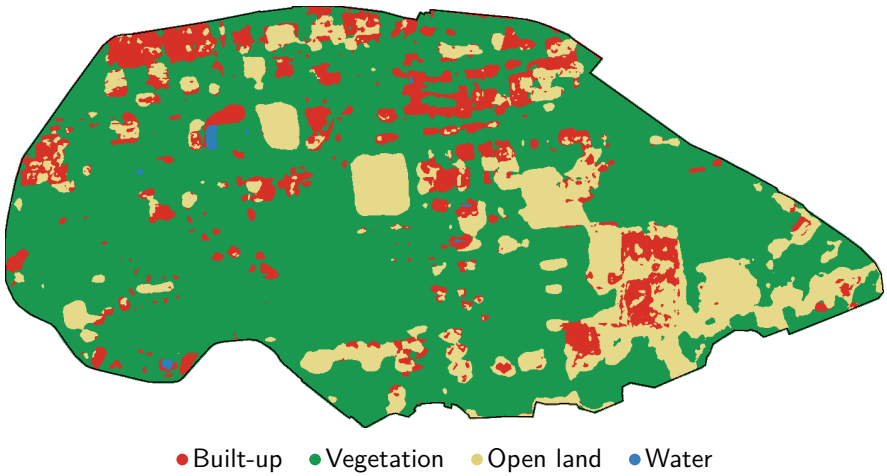
Land Use / Land Cover Mapping and Change Detection

IIT Kharagpur campus, 2020 vs 2025, from multi-temporal Sentinel-2

Ashutosh Sharma 23CS10005
Krishnkant Sahu 23CS10035
Sanskar Sovitkar 24CS10131

Term project evaluation, Nov 2026

2025 land cover map, Random Forest, 10 m



Can we measure how our own campus changed?

RESEARCH QUESTION

How much of the 469 ha campus changed land cover between the 2019-20 and 2024-25 dry seasons, and from what to what?

Why a campus? Global products (Dynamic World, ESA WorldCover) map at 10 m but are tuned for regions. A 469 ha site tests whether they hold at the scale where planners actually decide: one construction compound, one lawn, one pond.

01 **Map 4 classes at 2 dates**
Sentinel-2, one Random Forest

02 **From-to change table in ha**
30 m minimum change width

03 **Accuracy against independent points**
WorldView-2 and -3, area-weighted

04 **Reproducible pipeline**
one Earth Engine script, one Python runner

New in this version: is the change real? error-adjusted significance test · change flow diagram · gross gain, loss and swap per class · map area vs point-based area with 95% CI · model tuning log with a held-out half · feature importance · WorldView validation design

A 469 ha fenced campus, mapped at 10 m



Campus boundary: OpenStreetMap way 52435606 (184 vertices). Imagery: Google Earth, Airbus.

469.2 ha
MAIN FENCED CAMPUS

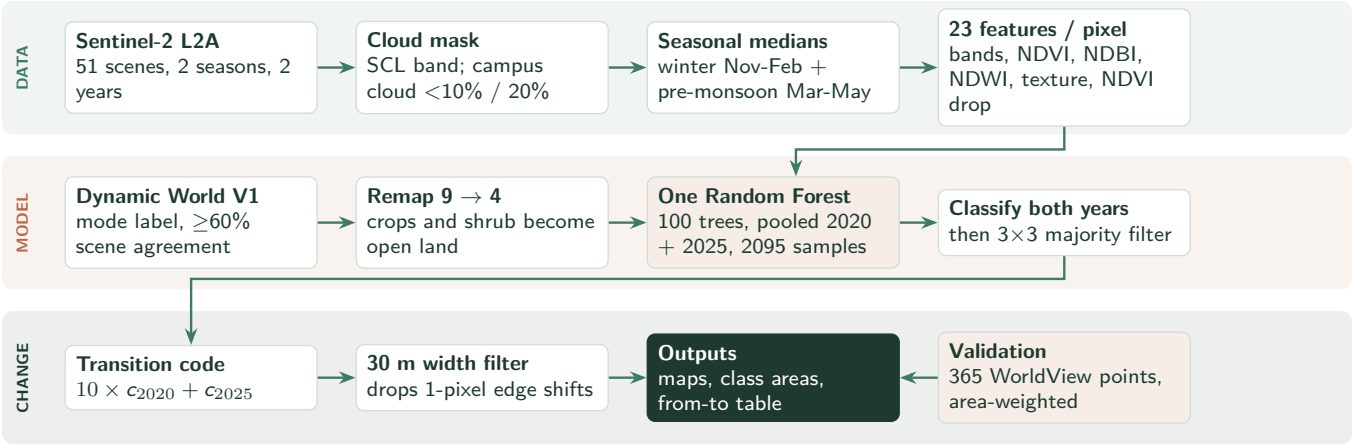
10 m
SENTINEL-2 PIXEL SIZE

45QWE
ONE TILE COVERS ALL OF IT

UTM 45N
EPSG:32645

Halls and quarters outside the fence are not in the study area.

From 51 satellite scenes to a from-to change table



One forest for both years, so an unchanged pixel gets the same rules in 2020 and 2025. Separate forests disagreed on 91 ha of unchanged roofs.

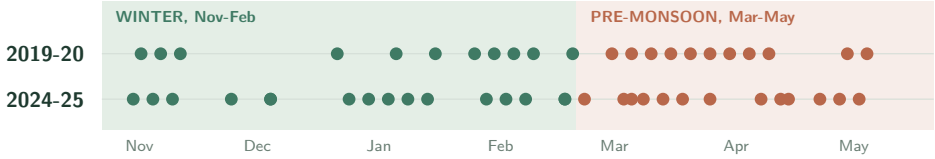
Same seasons in both years, so seasonal greening is not counted as change



Winter 2019-20 median composite, 11 scenes

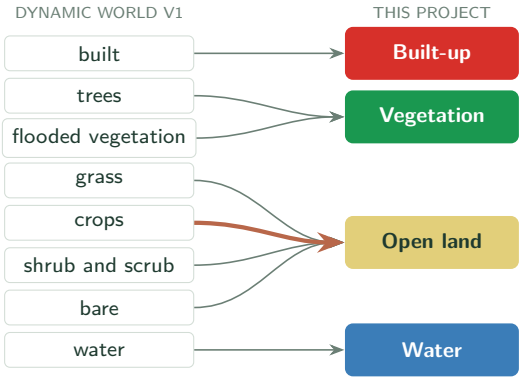


Winter 2024-25 median composite, 16 scenes

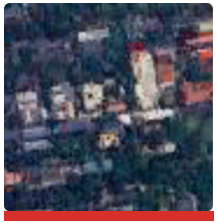


Every dot is one scene that passed the campus cloud test. By April lawns dry out but tree canopy stays green, so the pre-monsoon season helps split them.

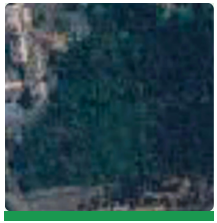
Nine Dynamic World labels collapse into four campus classes



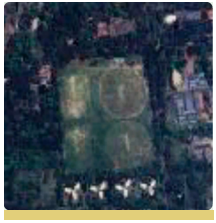
The crops rule. Dynamic World calls about 50 ha of campus lawns *crops*. There is no farmland inside the fence, so crops become open land. Calling them vegetation would hide open-to-built conversions.



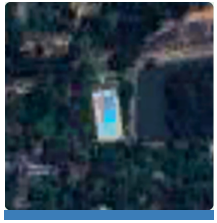
Built-up
roofs, roads, courts



Vegetation
canopy, dense woody



Open land
lawns, fields, bare plots



Water
ponds, tanks

23 features per pixel; the seasonal and texture ones earn their place

10	winter bands B2-B8A, B11, B12
3	winter NDVI, NDBI, NDWI
2	texture: 3×3 std of NDVI, B8
7	pre-monsoon: 4 bands, 3 indices
1	NDVI drop, winter minus pre-monsoon

= 23 features

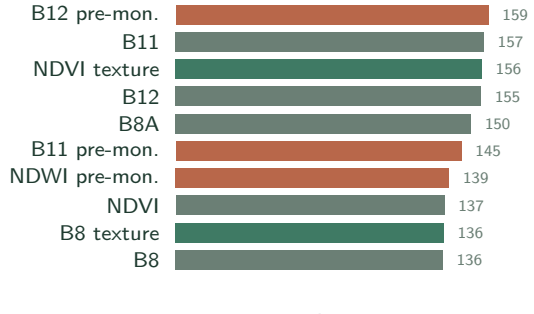
$$\mathrm{NDVI} = \frac{B8-B4}{B8+B4}$$

$$\mathrm{NDBI} = \frac{B11-B8}{B11+B8}$$

$$\mathrm{NDWI} = \frac{B3-B8}{B3+B8}$$

Greenness, built-up and water, each separating a pair of classes that look alike in raw bands.

RANDOM FOREST IMPORTANCE, TOP 10 OF 23 (GINI, EARTH ENGINE)



The **top feature is pre-monsoon SWIR (B12)**, and **5 of the top 10** are seasonal or texture features added in the last fix. Short-wave infrared is bright on roofs and dry soil, dark on moist leaves.

Tuning the model: three runs, each fix aimed at one measured error

RUN 1

Baseline

Equal samples per class, one forest per year.

Errors: built-up 224 ha (Dynamic World says ≈110); 91 ha built → vegetation on unchanged roofs; zero open-land samples in 2020.

→

RUN 2

Fix the sampling

Samples near real class shares (300 / 700 / 450 / 40); **one forest pooled over both years**; crops → open land; 30 m change filter.

Left over: open land only 32 ha; points say about 146.

→

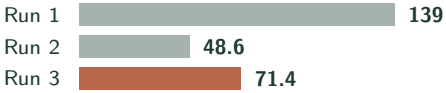
RUN 3 (FINAL)

Fix open land

Add pre-monsoon composite, 3×3 texture, shrub → open land, 450 open samples.

Result: open land 2020 at 103 ha; OA 0.61 → 0.67 (2020), 0.74 → 0.77 (2025).

CHANGED AREA REPORTED (HA)

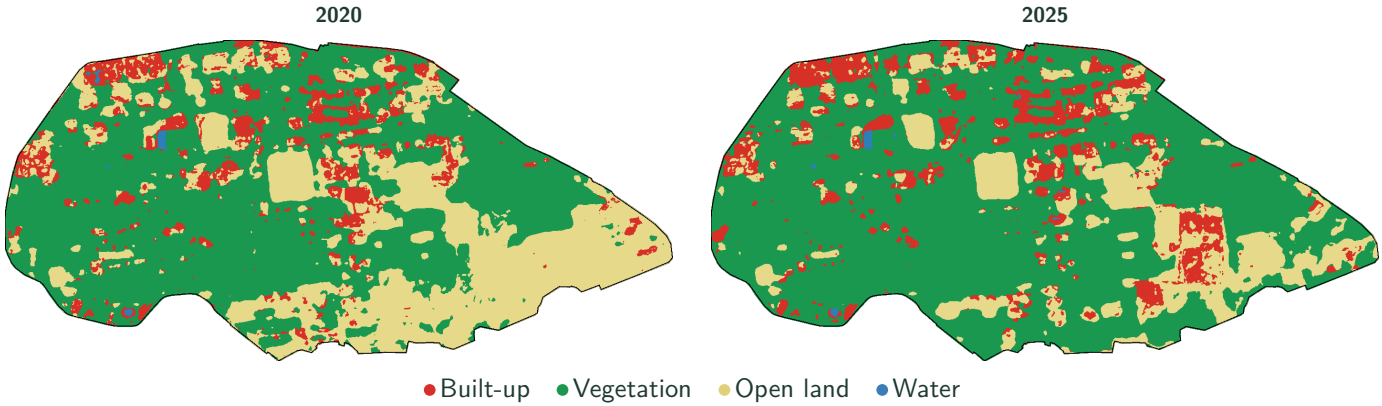


Held-out check, so tuning does not overfit the points. Each change was chosen on the odd-numbered validation points and scored on the even-numbered half it never saw:

2020 0.65 → **0.685** **2025** 0.70 → **0.76**

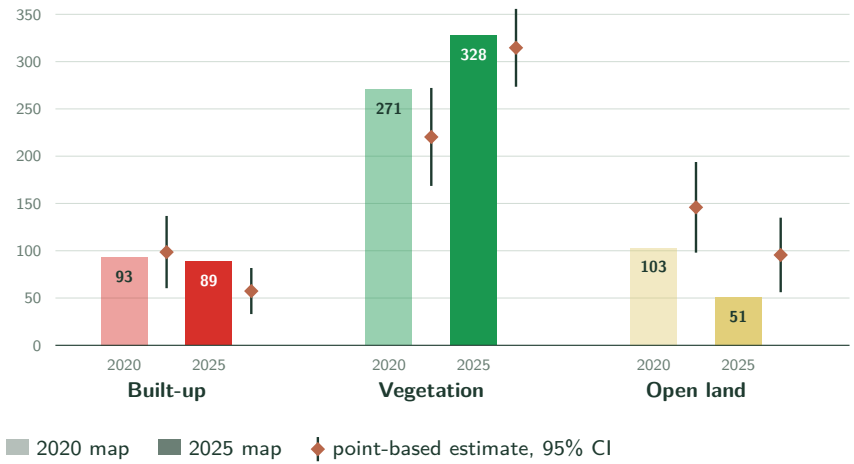
Run 2 to 3 raises change from 48.6 to 71.4 ha because more open land is now mapped, so more of it can be seen turning green.

Vegetation grew and open land shrank in five years



Open land (tan) in the east and south-east turns green or red. Same forest, same 23 features, both years, so the difference is the ground, not the model.

Vegetation +42.2 ha, open land −46.8 ha; the points agree on direction



Vegetation +42.2
gained 48.9, lost 6.7 ha

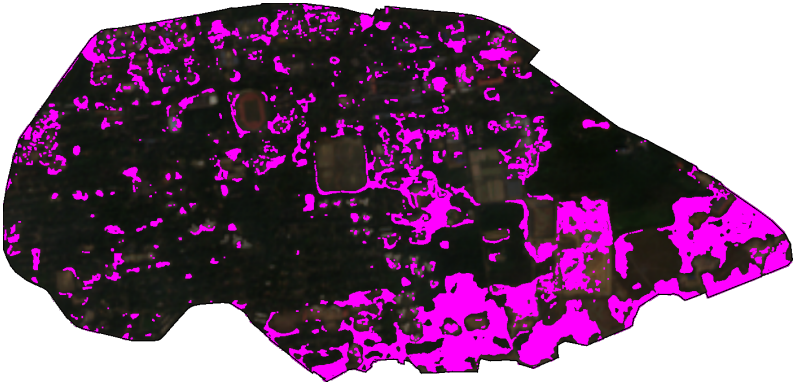
Open land −46.8
gained 3.4, lost 50.2 ha

Built-up +4.6
gained 19.0, lost 14.5 ha

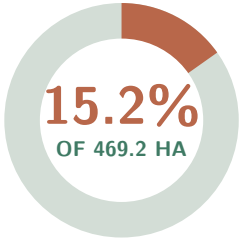
Cards: net after the 30 m filter. Bars also count the roof-edge ring it drops (9.4 ha of built-up).

Map vs points. Pixel counting under-counts open land by about 45 ha in both years. The point estimate (Olofsson et al. 2014) corrects for known errors; only vegetation's intervals do not overlap (169-272 vs 273-356 ha).

15.2% of the campus changed land cover in five years



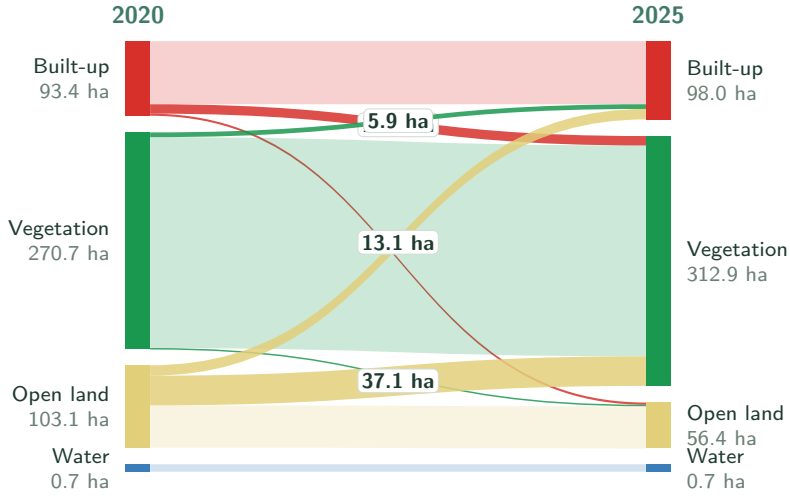
Magenta: changed patches at least 30 m wide, over the 2025 image



71.4 ha
CHANGED
≈14 ha / year

Before the 30 m width filter the change map holds 103.3 ha. The 31.9 ha it removes is mostly a one-pixel ring around every roof, where the two composites shift by a pixel.

Where the change went: open land feeds both vegetation and building



Band width = hectares. Faint bands stayed the same class; strong bands changed. From-to table after the 30 m filter.

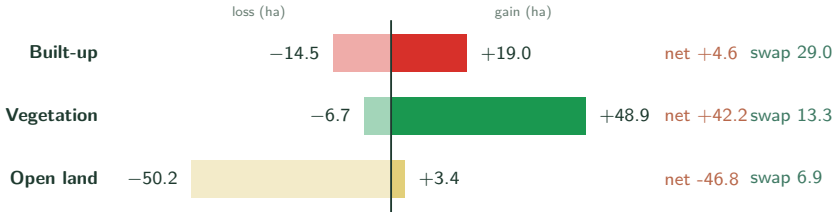
37.1 ha
Open → vegetation
east-end grassland growing into scrub

13.1 ha
Open → built-up
south-east construction compound

11.8 ha
Built-up → vegetation
mostly canopy closing over roofs

5.9 ha
Vegetation → built-up
new blocks in the south

Net change hides most of the built-up story



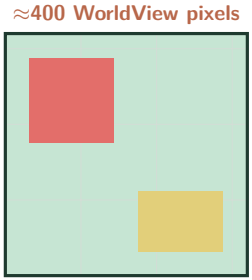
Hectares from the from-to table. Swap = 2 × min(gain, loss): area that changed place without changing the class total (Pontius et al. 2004).

Built-up looks stable. Net change is under 5 ha, yet 33.5 ha of built-up pixels changed: 19.0 ha gained, 14.5 ha lost. **Most of its change is swap.**

Vegetation and open land are one-way. Vegetation gains 48.9 and loses 6.7; open land loses 50.2 and gains 3.4. This is real, directional conversion.

Caveat: part of built-up loss is canopy closing over roofs, so some of the swap is the 10 m pixel, not new ground.

Validation labels come from WorldView, 20 times sharper than Sentinel-2

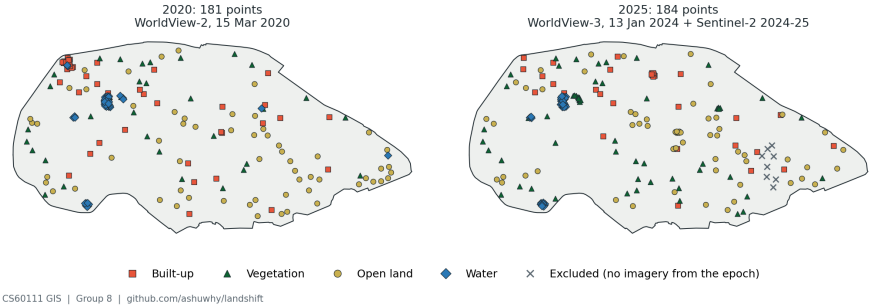


one Sentinel-2 pixel = 10 m
grid: 0.5 m, WorldView pan scale

A 10 m pixel that mixes a roof and a lawn is one label to the classifier but readable by eye on WorldView.

Stratify by map class, 25+ points per class per year → **Label by eye** WorldView-2 (15 Mar 2020), WorldView-3 (13 Jan 2024) → **Offset-correct** to the Sentinel-2 grid → **Area-weight** OA, user's, producer's, area ± 95% CI

Independent validation points, labelled on high-resolution imagery



365 points: 181 for 2020, 184 for 2025, labelled by Krishnkant Sahu. Independent of Dynamic World, unlike the 883 held-out training pixels.

Accuracy is 0.67 in 2020 and 0.77 in 2025, with wide intervals

OVERALL ACCURACY 2020

0.67

95% CI 0.57 to 0.78

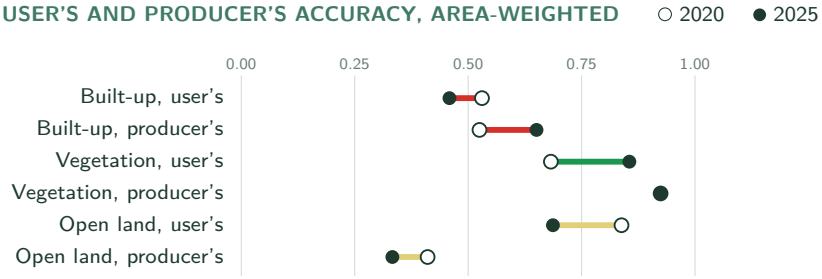
OVERALL ACCURACY 2025

0.77

95% CI 0.67 to 0.87

Against Dynamic World's own held-out pixels the map scores 0.82 / 0.88 (kappa 0.73 / 0.81), but that only measures agreement with the training source.

USER'S AND PRODUCER'S ACCURACY, AREA-WEIGHTED



Open land is the weak class: producer's 0.41 and 0.33, so about 45 ha of real open land is mapped as built-up or vegetation. Vegetation is found 92% of the time in both years.

Where the map goes wrong: open land read as built-up

2020: OA 0.67 (0.57 to 0.78)

reference (WorldView) \ map	Built	Veg	Open	Water
Built	32	5	5	2
Veg	4	27	1	2
Open	13	7	40	0
Water	9	9	0	25

2025: OA 0.77 (0.67 to 0.87)

reference (WorldView) \ map	Built	Veg	Open	Water
Built	23	3	5	0
Veg	3	52	2	1
Open	21	5	33	0
Water	3	2	0	31

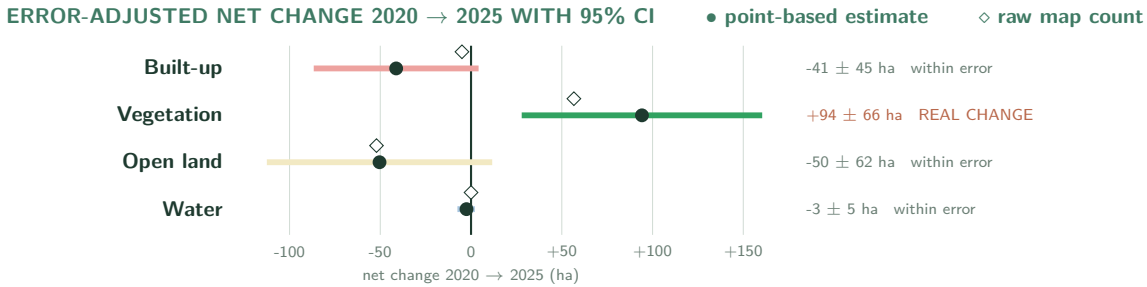
Rows: WorldView reference. Columns: map. Shade: share of the row. Green diagonal is correct; terracotta is error.

Open → built-up
13 of 60 points in 2020, 21 of 59 in 2025. Bare plots and dry courts share the SWIR signal of roofs.

Water → built-up or vegetation
18 of 43 water points in 2020 sit on pond edges, where a 10 m pixel is half bank.

Vegetation holds
52 of 58 in 2025. The forest is strongest on the class that dominates the campus.

Only the vegetation gain is bigger than our map error



Method
Olofsson adjusted area per class, per year, from the WorldView points. The 2020 and 2025 point sets are independent, so the half-width of the difference is $\sqrt{h_{2020}^2 + h_{2025}^2}$.

Vegetation $+94 \pm 66$ ha
The whole interval sits above zero ($z = 2.8$). Campus greening is the one change we can defend against map error.

Built-up and open land
Intervals cross zero (-41 ± 45, -50 ± 62 ha). Halving them needs about $4\times$ the points (CI $\propto 1/\sqrt{n}$), so that is where new labelling should go.

What the map can and cannot support

Open land is under-counted

Producer's 0.41 and 0.33; about 45 ha mapped as built-up or vegetation.

So: treat open-land area as a lower bound.

Change accuracy is not measured

The change map combines two maps at 0.67 and 0.77.

So: read 71.4 ha as an estimate.

10 m pixels mix surfaces

Roofs under canopy and small courts blur; B5-B7, B8A, B11, B12 are 20 m natively.

So: small features are unreliable.

The 30 m filter drops narrow change

103.3 $\rightarrow$ 71.4 ha; class totals count every pixel, the from-to table does not.

So: the two need not reconcile.

Labels come from Dynamic World

Lawns are called crops (50 ha in 2020); we remap by rule.

So: independent points check that rule.

Few points, tiny water

CIs are about ± 0.10 wide; water is 0.7 ha and 18 of 43 points sit on pond edges.

So: water is not estimable.

Two dates only: we can say what changed, not when.

15.2% of the campus changed, mostly open land turning green

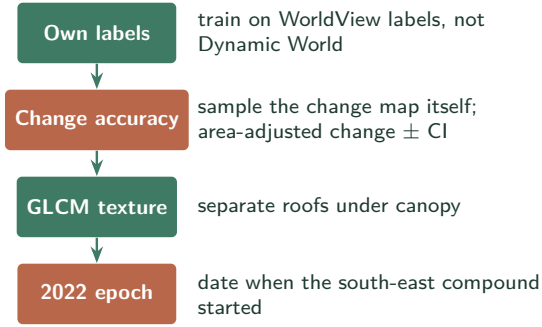
71.4 ha
changed: 37.1 open → vegetation, 13.1 open → built-up (south-east compound), 11.8 built-up → vegetation

+42.2 / −46.8 ha
net, after the 30 m filter: vegetation gained 48.9 and lost 6.7 ha; open land lost 50.2 and gained 3.4 ha

33.5 ha
of built-up changed hands while its net moved under 5 ha

0.67 / 0.77
overall accuracy on 365 independent WorldView points

WHAT WE WOULD DO NEXT



Who did what

Ashutosh Sharma
23CS10005

- Earth Engine pipeline, first drafts to `gee/lulc_pipeline.js`
- Diagnosed run 1 and ran the tuning: proportional sampling, one forest, crops rule, 30 m filter, pre-monsoon and texture
- Area-weighted accuracy and area estimates (Olofsson, Stehman)
- Python runner that reproduces every number; all figures, README and this deck

Krishnkant Sahu
23CS10035

- Independent validation set: 365 points, stratified on map classes
- Labelled on WorldView-2 and -3, offset-corrected to Sentinel-2
- Labelling notes: dates, rules, hard cases
- Point layers, sampling and export scripts

Sanskar Sovitkar
24CS10131

- Study-area figure: Google Earth Pro view with OSM boundary, north arrow, scale bar
- Literature review of Sentinel-2 / Random Forest change studies
- Framed the research gap: global products vs one campus

Code and results: **github.com/ashuwhy/landshift** · gee/lulc_pipeline.js · scripts/run_pipeline.py · slides/make_charts.py

Thank you

15.2% of the campus changed, mostly open land turning green.

github.com/ashuwhy/landshift

Ashutosh Sharma · Krishnkant Sahu · Sanskar Sovitkar

